

SECTION A

1 (**short**) Figure 1 shows a tall square closed can, with thin walls of uniform thickness t , subject to uniform external pressure P in excess of the internal pressure P_0 . The faces of the can remain planar under load.

Derive formulae for the stresses σ_l and σ_c in the vertical wall shown in the figure, stating any assumptions you make.

[10]

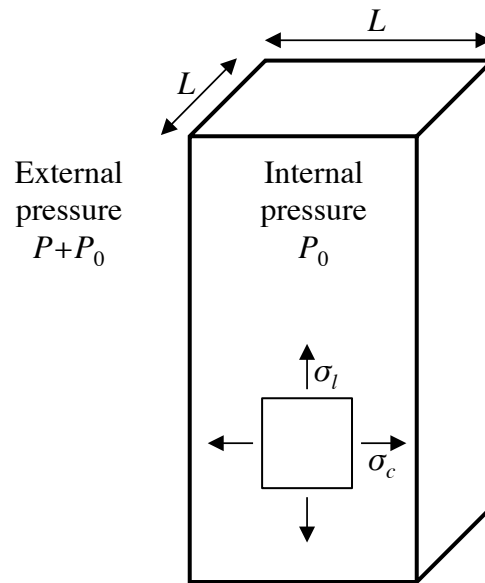


Fig. 1